TEXORA · LECTURE SHEET

Shuttleless Weaving Machines

Five ways to throw a pick without a shuttle: a steel projectile, two rapier hands, a jet of air, a jet of water, and a shed that opens in waves so several picks fly at once. This sheet studies each insertion system in detail, then compares all five on the only fair scale — weft insertion rate and metres per shift, calculated.

FM-4101 · ADVANCED WEAVING TECHNOLOGY · TOPIC 2 · SPECIALIZATION — FABRIC MANUFACTURING

1. Why the shuttle had to go

The shuttle carries its weft supply through the shed: it is heavy, it must be accelerated and stopped every pick, and its pirn limits both speed and unattended running. Every shuttleless system separates the two jobs — the weft is stored on a stationary cone outside the loom and measured by an accumulator, while a light carrier (or a fluid) takes a single pick across. The comparison metric follows at once:

$$\text{Weft insertion rate, WIR (m/min)} = \text{ppm} \times \text{reed width (m)}$$

A loom's ppm alone says nothing across widths; WIR is the speed at which yarn is actually consumed and cloth actually made.

2. Projectile weaving

A small steel projectile (about 40–90 g) grips the weft tip and is shot across the shed by a torsion-bar picking mechanism, guided by a row of guide teeth; it is braked at the receiving side, the yarn released, and the projectile returned under the machine by conveyor. One projectile crosses at a time, but several circulate. **Strengths:** the widest working widths of any system (up to about 5.4 m), gentle yarn handling that suits delicate and coarse yarns alike, excellent for industrial and heavy fabrics. **Limits:** moderate ppm; the picking energy is spent on the projectile, not the yarn, so efficiency per pick is poor at narrow widths.

3. Rapier weaving

Two rapiers — flexible tapes or rigid rods — work as giver and taker: the giver carries the weft tip to mid-shed, transfers it to the taker, which draws it through. Transfer is positive: the yarn is held at every

instant, which makes the rapier the most **versatile** system — fancy, multi-colour (up to 8–16 colours by selector), delicate, heavy, filament or spun, all on one machine type. **Limits:** the double traverse (rapier travels across and back empty of progress for half its motion) caps WIR; machine footprint is wide because the tapes need room beyond the shed on both sides.

4. Air-jet weaving

A main nozzle's air blast accelerates the weft into a profiled reed channel; relay nozzles along the width sustain its flight; a stretching nozzle holds it until beat-up. Nothing mechanical crosses the shed — hence the highest ppm of the single-pick systems. **Strengths:** speed and simplicity of the insertion itself; the standard for high-volume light-to-medium fabrics (sheeting, shirting, filament linings). **Limits:** compressed air is the mill's most expensive energy — air consumption dominates its running cost; yarn must tolerate air drag (very hairy, very weak or very heavy yarns fly badly); width is limited by air decay, mitigated but not abolished by relay nozzles.

5. Water-jet weaving

The same single-jet principle with water: a precisely metered jet carries the pick, and the fabric is dried on the machine. Water is incompressible and coherent over distance, so insertion is positive and fast with very low energy cost per pick. **The absolute limit:** only hydrophobic filament yarns (polyester, nylon) can be woven — the warp is wet throughout weaving, so sized natural-fibre warps would dissolve their size and rot their strength. Within that kingdom — filament taffeta, lining, umbrella cloth — it is dominant.

6. Multiphase weaving

The shed itself is rethought: instead of one shed opening across the full width, a wave of partial sheds travels along the warp (rotary or linear shed-wave), each carrying its own pick — several insertions are in flight simultaneously and beat-up is continuous by a rotating reed. Total insertion is the sum over simultaneous picks, reaching equivalent WIR values far beyond any single-pick machine. **Limits:** pattern flexibility is restricted (plain and simple constructions dominate), the mechanism is complex, and the principle has remained a specialist rather than a universal answer — a standing lesson that insertion rate is not the only term in the weaving equation.

7. The comparison, calculated

SYSTEM	PPM	REED WIDTH (M)	WIR (M/MIN)	M/SHIFT AT 70 PPI*
Projectile	260	3.90	1,014	38.5 (at 85%)

Rapier	550	1.90	1,045	84.3 (at 88%)
Air-jet	1,100	1.90	2,090	172.4 (at 90%)
Water-jet	1,200	1.90	2,280	192.3 (at 92%)
Multiphase	multi-pick	1.90	≈6,840 equivalent†	— (construction-limited)

* Metres per 8-hour shift by Sheet 4's formula, at each system's typical working efficiency. †Illustrative equivalent: simultaneous picks summing to ≈3,600 picks/min across 1.9 m. Efficiencies differ because stoppage causes differ — filament systems stop less on weft breaks; air-jet stops include air-supply events. Widths differ because that is each system's home ground — the projectile's 3.9 m is the point of it.

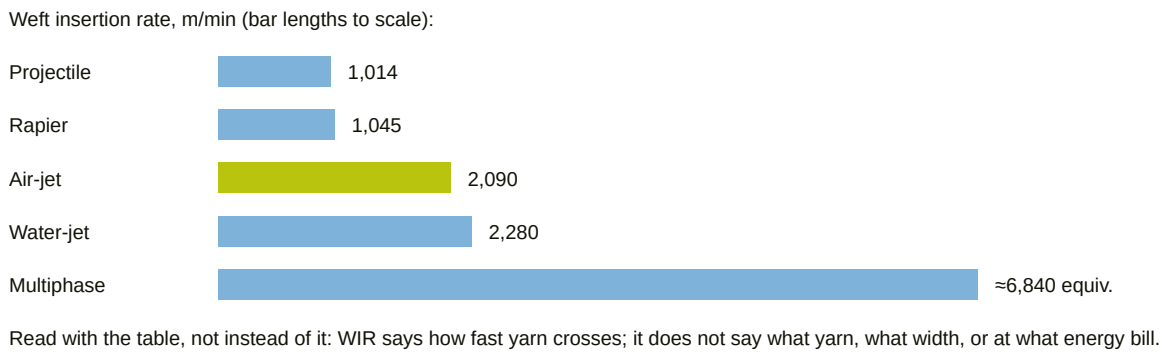


Figure 1 — Weft insertion rates of the five systems on their home widths.

8. Choosing the system

DECIDING FACTOR	POINTS TO
Very wide or heavy/industrial fabric	Projectile
Many weft colours, fancy, frequent style change	Rapier
Volume light/medium fabrics, spun or filament	Air-jet
Hydrophobic filament only, lowest energy per pick	Water-jet
Extreme insertion rate, simple constructions	Multiphase
Selvedge	All form tucked, leno or fused selvedges — the shuttle's woven edge is gone everywhere; waste selvedge (fringe) on jet looms must be costed and

recycled

Weft supply closes the calculation: at 70 PPI and 1.9 m with 3% waste allowance, one metre of fabric eats 70 × 39.37 × 1.9 × 1.03 ≈ **5.39 km of weft** — at air-jet rate, the accumulator and creel of a single loom draw yarn at over 2 km per minute. Insertion systems are chosen by fabric; they are fed by logistics.

Glossary

WIR — weft insertion rate, ppm × reed width. **Accumulator (weft feeder)** — measures and stores a pick length, releasing it on demand. **Relay nozzles** — auxiliary air jets sustaining weft flight across the shed. **Shed wave** — multiphase's travelling partial sheds. **Tucked/leno/fused selvedge** — the shuttleless edge finishes replacing the shuttle's folded edge.

Key points

• Compare looms by WIR = ppm × width, never by ppm alone (worked table: 1,014 / 1,045 / 2,090 / 2,280 m/min).
• Each system's strength is a fabric statement: width (projectile), versatility (rapier), volume (air-jet), filament economy (water-jet), raw rate (multiphase).
• Each limit is physical: rapier's empty return, air's energy cost and drag, water's hydrophobe-only rule, multiphase's construction ceiling.
• Metres per shift still come from Sheet 4's formula — the systems differ inside its terms (ppm, efficiency), not outside them.

Practice

1. An air-jet loom is offered at 1,400 ppm on 1.5 m width against a rapier at 600 ppm on 2.2 m. Compare their WIRs and state what the comparison omits.
2. Why can a water-jet loom never weave the FM-3207 reference cotton fabric, however attractive its WIR?
3. Using the table, how many air-jet looms (172.4 m/shift) match the daily output of 20 projectile looms on the same fabric and shift pattern?

Answer key

1. Air-jet WIR = 1,400 × 1.5 = **2,100 m/min**; rapier = 600 × 2.2 = **1,320 m/min**. The air-jet wins the rate — but the comparison omits width itself (the rapier's cloth is wider per metre of output), colour capability, yarn suitability and the air bill.

2. Cotton warp must carry water-soluble size and is weakened and rotted by continuous wetting; the water jet would strip the size and destroy the warp's weaving strength within metres. The system's kingdom is hydrophobic filament by physics, not by preference.
3. 20 projectiles × 38.5 = 770 m/shift; 770 ÷ 172.4 = 4.5 → **5 air-jet looms** — on this fabric and width class; change the fabric and the answer is rebuilt, not reused.

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